Marmara University

Electrical and Electronics Engineering Department

Fall 2023

EE 3016/3051 – Electromagnetics I

Simulation Project - Due December 22, 2023

Prepared By

Yunus Emre Turunç-150721854

Mehmet Ali Kamalak-150720062

Ebubekir Mutugan-150720002

Ömer Faruk Akyürek-150718035

Numerical Solution of Poisson's Equation

The main purpose of this simulation project is to numerically solve a boundary-value problem involving the electric potential. The electric potential distribution inside a given structures is obtained numerically. The finite-difference (FD) method is used to solve Poisson's equation in two dimensions

Introduction

The electric potential distribution inside any given structure can be analytically obtained by solving Poisson's equation subject to some boundary conditions. If we assume no volume charge inside the structure, Poisson's equation becomes (Laplace's equation)

$$\nabla^2 V = 0 \tag{1}$$

In two dimensions (rectangular coordinates), equation (1) is written as

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial x^2} = 0 \tag{2}$$

The ability to solve equation (2) depends in a great deal on the nature of the structure under consideration. In some cases a closed-form (analytical) solution to equation (2) is difficult to obtain. Alternatively, numerical methods can be used, especially for large problems. Numerical methods utilize the speed of computers and the flexibility of programming.

In this project, the Finite-Difference (FD) method will be used. The FD method is one of the most popular numerical methods in the field of electromagnetics. Its main advantages are (1) easy to formulate, (2) suitable for many structures and (3) flexible for modifications.

The FD-based solution is performed in the following steps.

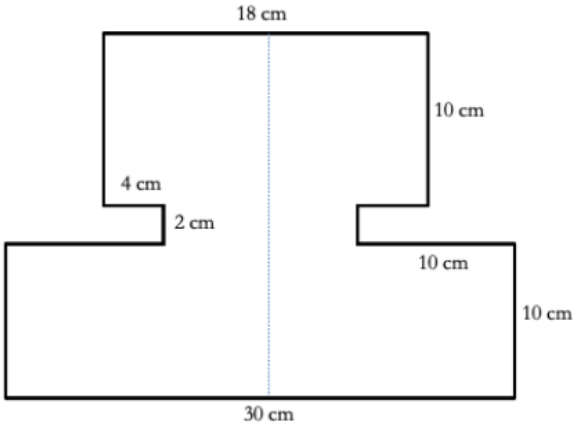
1) Discretize the given structure (make a grid) using a suitable step size in both dimensions (Δx and Δy). The accuracy of the results improves with smaller step sizes.
2) Approximate the partial derivatives in equation (2) by the following second-order relations.

$$\frac{\partial^2 V}{\partial x^2} = \frac{V(i+1,j) - 2V(i,j) + V(i-1,j)}{\Delta x^2} \tag{3}$$

$$\frac{\partial^2 V}{\partial y^2} = \frac{V(i,j+1) - 2V(i,j) + V(i,j-1)}{\Delta y^2} \tag{4}$$

where i and j are the indices along the x- and the y-axis, respectively. Substituting equations (3) and (4) in equation (2), and using $\Delta x = \Delta y$, we get

$$V(i,j) = \frac{1}{4}[V(i+1,j) + V(i-1,j) + V(i,j+1) + V(i,j-1)] \tag{5}$$



The boundary of the left half of the structure is biased at -1V, whereas the boundary of the right half is biased at +5V. Use a step size of 2 cm in both dimensions.

According to these procedures, first we set the boundaries for 2 cm step size.

Part (1)

```
% Parameters
lengthX = 30; % cm
lengthY = 30; % cm
n4=2; %step size

% % Discretize the grid
y = 0:n4:lengthY;
x = 0:n4:lengthX;

% Initialize potential matrix
stepsize2 = zeros(length(x),
length(y));

% Set boundary conditions
(other points must be zero)

%top of the figure
for i = 4:8
    stepsize2(5,i) = -1;
end

%top of the figure
for a = 9:13
    stepsize2(5,a) = 5;
end

%left and right side of the
figure
for b = 6:9
    stepsize2(b,4) = -1;
    stepsize2(b,13) = 5;
end
```

Part (2)

```
%left side of the figure
stepsize2(10,4 : 6) = -1;
stepsize2(10,6) = -1;
stepsize2(11,1 : 6) = -1;

%right side of the figure
stepsize2(10,11 : 13) = 5;
stepsize2(10,11) = 5;
stepsize2(11,11 :16) = 5;

%left and right side of the
figure
for j = 12:15
    stepsize2(j,1) = -1;
    stepsize2(j,16) = 5;
end

%bottom side of the figure
for c = 1:8
    stepsize2(16,c) = -1;
end

%bottom side of the figure
for d = 9:16
    stepsize2(16,d) = 5;
end
```

Then, we give into the formula for potential values using the Finite-Difference formula.

```
% Iterative solution using Finite-Difference method
maxIterations = 1000;
tolerance = 1e-10;

for repeat = 1:10
    for iteration = 1:maxIterations

        % Update potential values using the Finite-Difference formula
        V_old = stepsize2;

        for i = 6:9
            for j = 5:12
                stepsize2(i, j) = ((stepsize2(i+1, j) + stepsize2(i-1, j) + stepsize2(i, j+1) + stepsize2(i, j-1)))/4;
            end
        end
         % Check for convergence
        if max(abs(stepsize2 - V_old)) < tolerance
            break;
        end
    end
    for iteration = 1:maxIterations

        % Update potential values using the Finite-Difference formula
        V_old = stepsize2;
        for t = 10:11
            for y = 7:10
                stepsize2(t, y) = ((stepsize2(t+1, y) + stepsize2(t-1, y) + stepsize2(t, y+1) + stepsize2(t, y-1)))/4;
            end
        end
         % Check for convergence
        if max(abs(stepsize2 - V_old)) < tolerance
            break;
        end
    end
        % Update potential values using the Finite-Difference formula
        V_old = stepsize2;
    for iteration = 1:maxIterations

        for o = 12:15
            for p = 2:15
                stepsize2(o, p) = ((stepsize2(o+1, p) + stepsize2(o-1, p) + stepsize2(o, p+1) + stepsize2(o, p-1)))/4;
            end
        end
         % Check for convergence
        if max(abs(stepsize2 - V_old)) < tolerance
            break;
        end

    end
end
```

maxIterations and **tolerance** part iteratively updates the potential values in the matrix using the Finite-Difference method until convergence or a maximum number of iterations is reached.

Part (1)

```
% Display the results

disp('Electric Potential Distribution:');
disp(stepsize2);

% Plot the electric potential distribution
using contourf
figure;
contourf(stepsize2,50,'LineColor', 'none');

% Add labels and title
xlabel('X-axis (cm)');
ylabel('Y-axis (cm)');
title('Electric Potential Distribution Step
Size 2');

% Add colorbar
colorbar;

% Display grid lines
grid on;

% Show the figure
axis equal;
axis([0 lengthX 0 lengthY]);

% Optional: Customize colormap
colormap('jet');
```

Part (2)

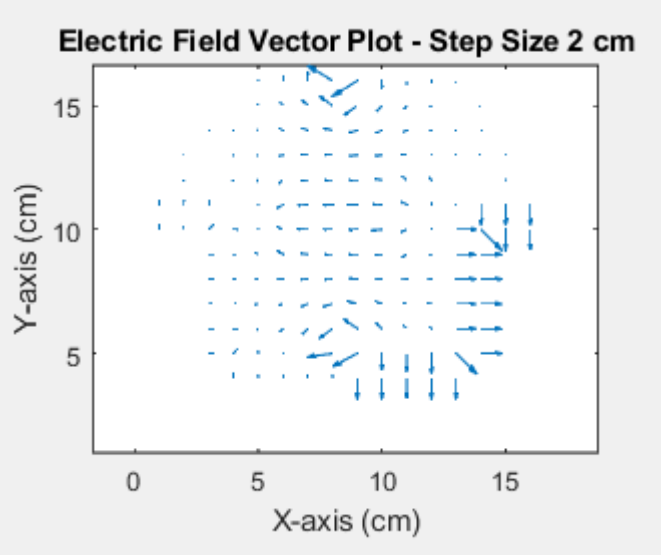
```
% Calculate electric field
components using central
differences

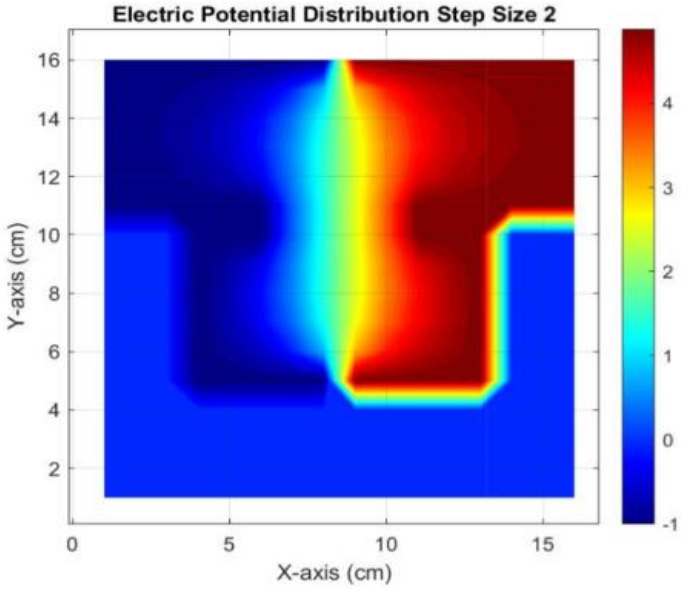
[Ex, Ey] = gradient(-
stepsize2);

% Define the grid for the
vector plot
[X, Y] = meshgrid(x, y);

% Plotting the vector field
figure;
quiver(X, Y, Ex, Ey);
title('Electric Field Vector
Plot Step Size 1 cm');
xlabel('X-axis (cm)');
ylabel('Y-axis (cm)');
axis equal;
```

This part displays the Electric Potential Distribution and Electric Field Vector.





Voltage Values in the structure for 2 cm step size.

			-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	5.0000	5.0000	5.0000	5.0000	5.0000		
			-1.0000	-0.8011	-0.5296	-0.0703	0.8548	3.1449	4.0700	4.5294	4.8010	5.0000		
			-1.0000	-0.6748	-0.2470	0.3936	1.3445	2.6547	3.6058	4.2466	4.6746	5.0000		
			-1.0000	-0.6511	-0.1772	0.5472	1.4749	2.5238	3.4518	4.1767	4.6508	5.0000		
			-1.0000	-0.7523	-0.3580	0.4976	1.4839	2.5138	3.5009	4.3574	4.7521	5.0000		
			-1.0000	-1.0000	-1.0000	0.3178	1.4505	2.5478	3.6812	5.0000	5.0000	5.0000		
-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	0.3229	1.4526	2.5457	3.6760	5.0000	5.0000	5.0000	5.0000	5.0000	5.0000
-1.0000	-0.9649	-0.9154	-0.8292	-0.6627	-0.3110	0.5218	1.4919	2.5071	3.4775	4.3107	4.6626	4.8292	4.9153	4.9649	5.0000
-1.0000	-0.9444	-0.8673	-0.7389	-0.5106	-0.1031	0.5834	1.4860	2.5134	3.4162	4.1028	4.5104	4.7388	4.8673	4.9444	5.0000
-1.0000	-0.9454	-0.8705	-0.7484	-0.5377	-0.1742	0.4289	1.3552	2.6445	3.5708	4.1741	4.5375	4.7484	4.8705	4.9454	5.0000
-1.0000	-0.9666	-0.9209	-0.8467	-0.7174	-0.4851	-0.0487	0.8613	3.1386	4.0486	4.4850	4.7173	4.8466	4.9209	4.9666	5.0000
-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	5.0000	5.0000	5.0000	5.0000	5.0000	5.0000	5.0000	5.0000

Then, we decrease the step size to 1 cm.

Part (1)

```
% Parameters
lengthX = 30; % cm
lengthY = 30; % cm
n=1;%choose step size

% % Discretize the grid
y = 0:n:lengthY;
x = 0:n:lengthX;

% Initialize potential matrix
stepsize1 = zeros(length(x),
length(y));

% Set boundary conditions
(other points must be zero)

%top of the figure
for i = 7:15
    stepsize1(9,i) = -1;
end

%top of the figure
for a = 16:25
    stepsize1(9,a) = 5;
end

%left and right side of the
figure
for b = 9:17
    stepsize1(b,7) = -1;
    stepsize1(b,25) = 5;
end

%left side of the figure
stepsize1(19,7 : 10) = -1;
stepsize1(18,7) = -1;
stepsize1(19,11) = -1;
stepsize1(20,11) = -1;
stepsize1(21,1 : 11) = -1;

%right side of the figure
stepsize1(19,21 : 25) = 5;
stepsize1(18,25) = 5;
stepsize1(19,21) = 5;
stepsize1(20,21) = 5;
stepsize1(21,21 :31) = 5;

%left and right side of the
figure
for j = 22:30
    stepsize1(j,1) = -1;
    stepsize1(j,31) = 5;
end

%bottom side of the figure
for c = 1:15
    stepsize1(31,c) = -1;
end

%bottom side of the figure
for d = 16:31
    stepsize1(31,d) = 5;
end
```

Part (2)

```
% Iterative solution using Finite-Difference method

maxIterations = 1000;
tolerance = 1e-10;

for repeat = 1:10
    for iteration = 1:maxIterations

        % Update potential values using the Finite-Difference formula
        V_old = stepsize1;

        for i = 10:18
            for j = 8:24
                stepsize1(i, j) = ((stepsize1(i+1, j) + stepsize1(i-
1, j) + stepsize1(i, j+1) + stepsize1(i, j-1)))/4;
            end
        end
         % Check for convergence
        if max(abs(stepsize1 - V_old)) < tolerance
            break;
        end
    end
    for iteration = 1:maxIterations

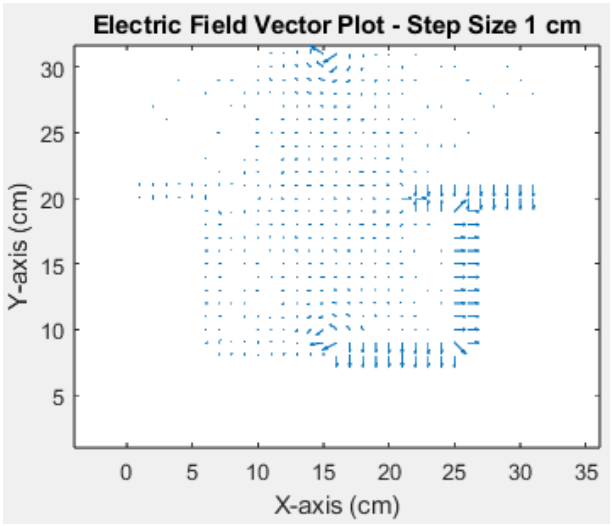
        % Update potential values using the Finite-Difference formula
        V_old = stepsize1;
        for t = 19:21
            for y = 12:20
                stepsize1(t, y) = ((stepsize1(t+1, y) + stepsize1(t-
1, y) + stepsize1(t, y+1) + stepsize1(t, y-1)))/4;
            end
        end
         % Check for convergence
        if max(abs(stepsize1 - V_old)) < tolerance
            break;
        end
    end
        % Update potential values using the Finite-Difference formula
        V_old = stepsize1;
    for iteration = 1:maxIterations

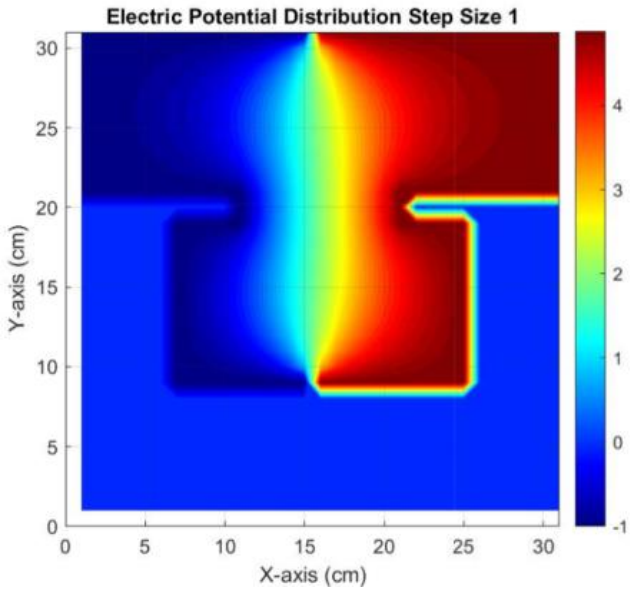
        for o = 22:30
            for p = 2:30
                stepsize1(o, p) = ((stepsize1(o+1, p) + stepsize1(o-
1, p) + stepsize1(o, p+1) + stepsize1(o, p-1)))/4;
            end
        end
         % Check for convergence
        if max(abs(stepsize1 - V_old)) < tolerance
            break;
        end
    end
end
disp('Electric Potential Distribution:');
disp(stepsize1);
figure;
contourf(stepsize1,50,'LineColor', 'none');
xlabel('X-axis (cm)');
ylabel('Y-axis (cm)');
title('Electric Potential Distribution');
colorbar;
grid on;
axis equal;
axis([0 lengthX 0 lengthY]);
colormap('jet');
[Ex, Ey] = gradient(-stepsize1);
[X, Y] = meshgrid(x, y);
figure;
quiver(X, Y, Ex, Ey);
title('Electric Field Vector Plot Step Size 1 cm');
xlabel('X-axis (cm)');
ylabel('Y-axis (cm)');
axis equal;
```

Voltage Values in the structure for 1 cm step size.

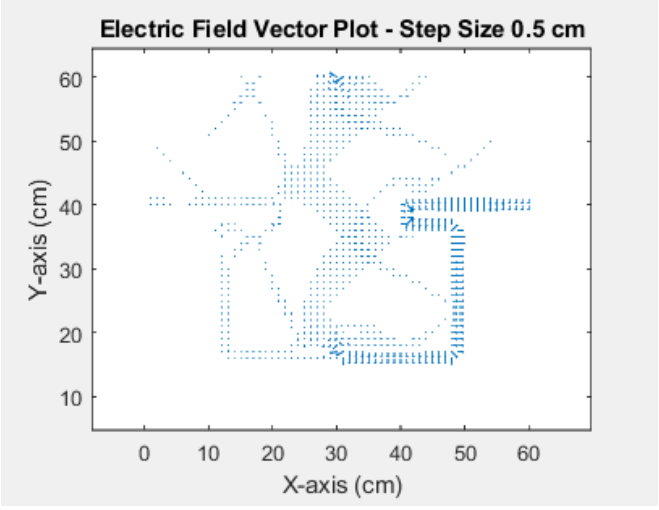
-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000
-1.000	-0.944	-0.883	-0.809	-0.713	-0.577	-0.362	0.029	0.874	3.081	3.927	4.319	4.535	4.673	4.770	4.843	4.902	4.953	5.000
-1.000	-0.894	-0.778	-0.641	-0.467	-0.234	0.100	0.603	1.387	2.522	3.308	3.813	4.150	4.387	4.562	4.700	4.812	4.910	5.000
-1.000	-0.853	-0.694	-0.509	-0.281	0.010	0.391	0.896	1.550	2.312	2.968	3.477	3.864	4.161	4.394	4.582	4.738	4.874	5.000
-1.000	-0.826	-0.637	-0.419	-0.157	0.164	0.560	1.041	1.603	2.210	2.776	3.263	3.668	3.999	4.271	4.495	4.683	4.847	5.000
-1.000	-0.812	-0.608	-0.374	-0.094	0.243	0.642	1.103	1.614	2.147	2.662	3.132	3.544	3.898	4.196	4.444	4.652	4.833	5.000
-1.000	-0.814	-0.611	-0.374	-0.086	0.260	0.664	1.115	1.602	2.103	2.594	3.057	3.479	3.852	4.171	4.433	4.649	4.832	5.000
-1.000	-0.833	-0.648	-0.423	-0.138	0.220	0.636	1.093	1.575	2.067	2.555	3.024	3.464	3.861	4.201	4.468	4.677	4.848	5.000
-1.000	-0.871	-0.724	-0.533	-0.264	0.121	0.569	1.046	1.539	2.037	2.533	3.020	3.490	3.928	4.304	4.563	4.744	4.881	5.000
-1.000	-0.929	-0.843	-0.720	-0.506	-0.039	0.471	0.984	1.497	2.008	2.521	3.034	3.548	4.058	4.524	4.735	4.853	4.934	5.000
-1.000	-1.000	-1.000	-1.000	-1.000	-0.232	0.387	0.947	1.482	2.009	2.535	3.070	3.629	4.242	5.000	5.000	5.000	5.000	5.000
				-1.000	-0.274	0.360	0.934	1.478	2.009	2.539	3.082	3.654	4.283	5.000				

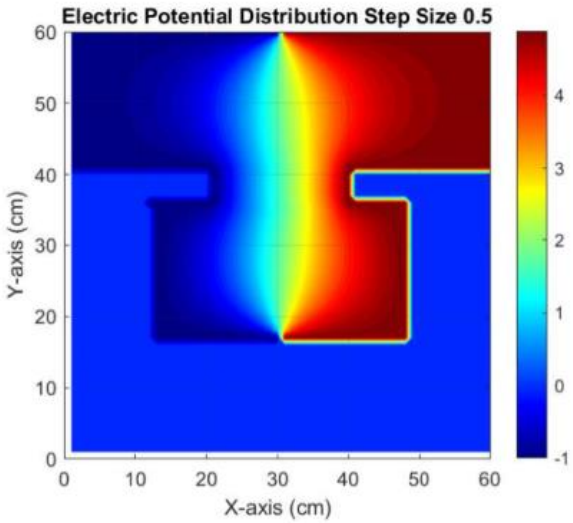
-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-0.224	0.394	0.952	1.485	2.009	2.533	3.065	3.622	4.236	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000
-1.000	-0.991	-0.980	-0.968	-0.952	-0.930	-0.901	-0.857	-0.791	-0.681	-0.474	-0.008	0.502	1.014	1.524	2.031	2.538	3.043	3.548	4.048	4.504	4.704	4.809	4.871	4.910	4.938	4.957	4.971	4.982	4.992	5.000
-1.000	-0.982	-0.963	-0.939	-0.909	-0.869	-0.814	-0.737	-0.626	-0.460	-0.208	0.167	0.606	1.078	1.564	2.055	2.543	3.021	3.480	3.905	4.265	4.504	4.660	4.763	4.834	4.883	4.919	4.946	4.967	4.984	5.000
-1.000	-0.976	-0.949	-0.917	-0.876	-0.823	-0.751	-0.652	-0.515	-0.325	-0.065	0.276	0.680	1.127	1.599	2.082	2.560	3.018	3.445	3.826	4.146	4.388	4.563	4.688	4.778	4.843	4.890	4.926	4.955	4.978	5.000
-1.000	-0.972	-0.940	-0.903	-0.856	-0.795	-0.714	-0.604	-0.457	-0.260	-0.002	0.323	0.710	1.149	1.625	2.115	2.596	3.047	3.454	3.810	4.106	4.339	4.517	4.649	4.747	4.819	4.873	4.915	4.947	4.975	5.000
-1.000	-0.970	-0.938	-0.899	-0.851	-0.788	-0.705	-0.595	-0.449	-0.256	-0.007	0.307	0.689	1.136	1.635	2.157	2.661	3.118	3.516	3.852	4.128	4.346	4.515	4.644	4.741	4.813	4.869	4.911	4.945	4.974	5.000
-1.000	-0.972	-0.941	-0.905	-0.859	-0.801	-0.724	-0.622	-0.487	-0.309	-0.076	0.223	0.602	1.070	1.622	2.218	2.773	3.249	3.640	3.956	4.206	4.403	4.554	4.671	4.759	4.826	4.877	4.917	4.948	4.975	5.000
-1.000	-0.976	-0.950	-0.920	-0.881	-0.832	-0.767	-0.682	-0.567	-0.415	-0.212	0.060	0.427	0.920	1.565	2.318	2.965	3.465	3.840	4.124	4.339	4.504	4.629	4.725	4.798	4.854	4.896	4.930	4.956	4.979	5.000
-1.000	-0.983	-0.964	-0.942	-0.914	-0.879	-0.832	-0.770	-0.686	-0.573	-0.418	-0.198	0.125	0.620	1.398	2.526	3.305	3.804	4.132	4.359	4.523	4.643	4.734	4.803	4.855	4.895	4.925	4.949	4.969	4.985	5.000
-1.000	-0.991	-0.981	-0.969	-0.955	-0.936	-0.912	-0.879	-0.835	-0.774	-0.688	-0.559	-0.349	0.038	0.879	3.083	3.925	4.314	4.526	4.659	4.749	4.813	4.861	4.897	4.924	4.945	4.961	4.973	4.984	4.992	5.000
-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	-1.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000

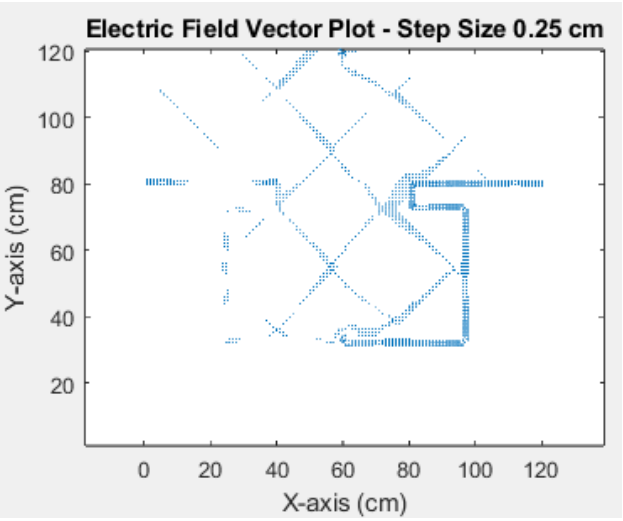


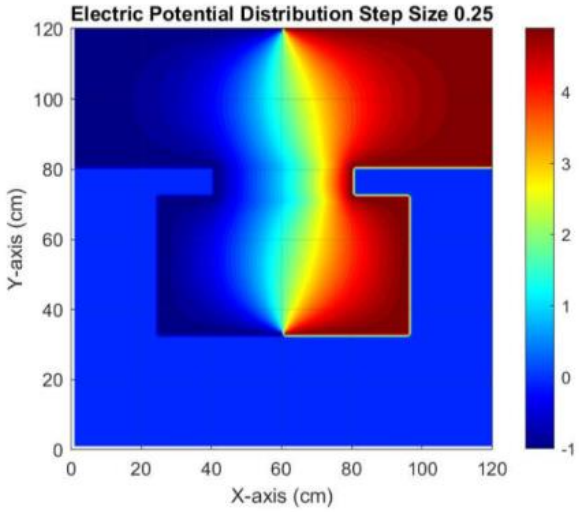


Then we repeat the same steps for step size 0.5 and 0.2.









Since the number of steps are 0.5 and 0.25, there are too many values, so we could not add the potential values here. You can see the values for 0.5 and 0.25 from the link below;

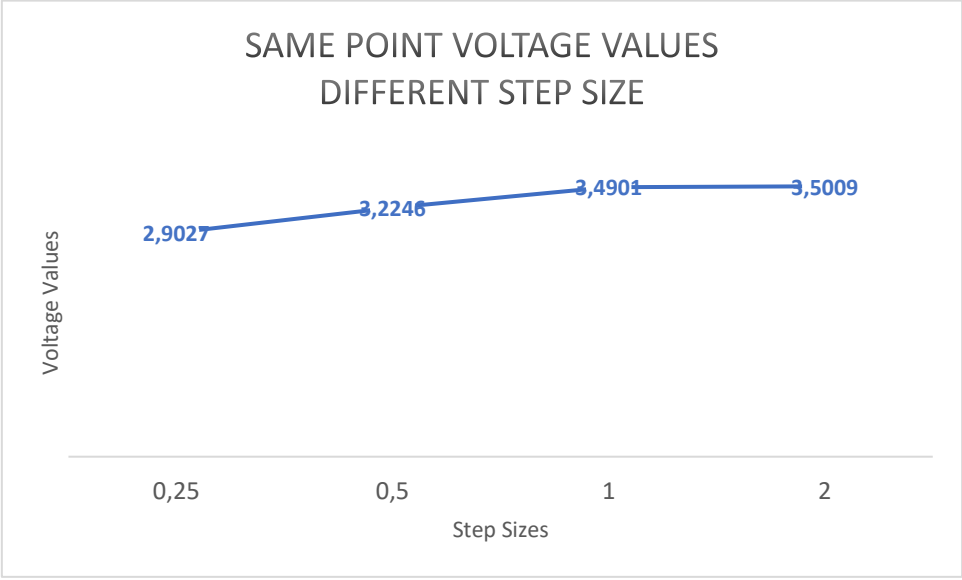
Step size 0.50: https://prnt.sc/q9QMwiuIAv_x

Step size 0.25: https://prnt.sc/6YgtReKJ-Bjv

ERROR ANALYSIS

If we think of the structure on the XY Plane. We can select one point for analysis. We select (16,12) point for observing on the change of voltage values.

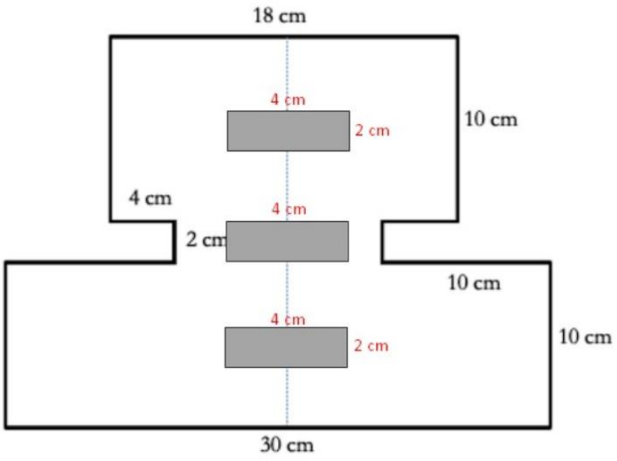
Step Sizes			
0.25	0.5	1	2
2.9027	3.2246	3.4901	3.5009



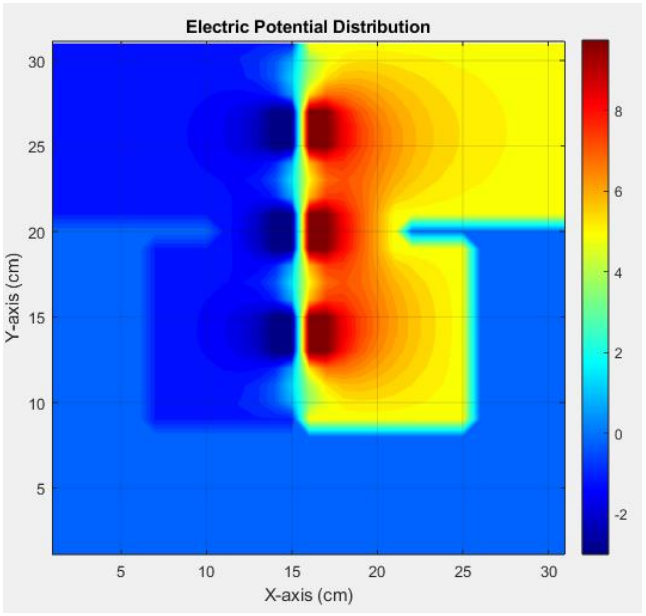


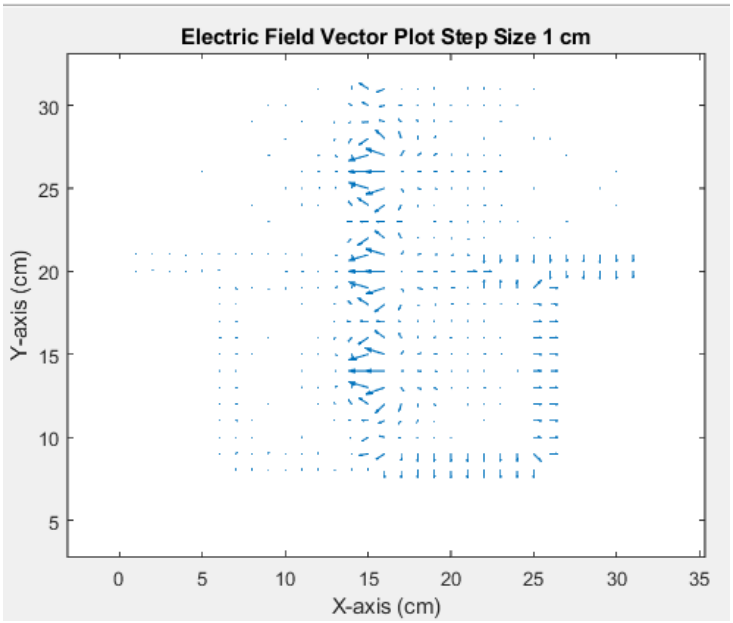
As a result, we find out the choice of step size affects the accuracy of the simulation. Smaller step sizes generally provide more accurate results but require more computational resources and time. Larger step sizes reduce computational requirements but may sacrifice accuracy, especially in capturing fine details of the potential distribution.

UNIQUE DESIGN



The boundary of the left half of the middle structures is biased at -3V, whereas the boundary of the right half is biased at +10V. We used step size of 1 cm for the finding new voltage points.





Voltage Values in the structure for 1 cm step size.

| | | |